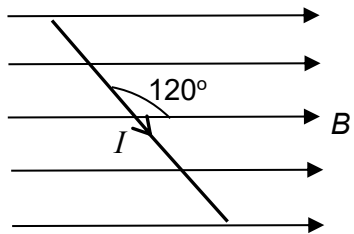
22 A wire 30 cm long with a mass of 4.0 g, is placed at an angle of 120° to a horizontal magnetic field of flux density 0.040 T. When a current I is passed through the wire, the wire accelerates uniformly upwards. The diagram below shows the top view of the set-up.



If the acceleration of the wire is 2.0 m s^{-2}, what is the current in the wire?

A 7.9 A **B** 4.5 A **C** 3.0 A **D** 0.77 A